

Dr Lawrence Yolland

SOFTWARE ENGINEER

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Researcher-turned-software engineer with a multidisciplinary background spanning computational biology, data analysis, and full-stack development. Experienced building high-quality, client-facing React applications, constructing APIs in Node.js, Go, and Ruby, and developing custom data-processing tools in Python and MATLAB. Equally comfortable working with research scientists, product teams, and enterprise stakeholders. Motivated by problems that sit at the intersection of rigorous analysis and practical engineering.

EXPERIENCE

Software Engineer

Red Bee Media | 2020 -

- Lead developer of a business-critical, customer-facing React dashboard with fully customisable tiles (data, size, position, colour, text), displaying workflow summary stats and detailed graphical renderings.
- Lead developer on a business-critical playout dashboard (used by operations teams responsible for broadcast transmission) across major broadcasters, displaying schedules and broadcast rundowns in real-time. Inherited from another team, as the new project owner I conducted a security audit, updated outdated dependencies, and modernised the codebase and developer workflow to align with our established technical stack.
- Developed a full-stack internal tool in TypeScript (React frontend, Node.js/Express backend) for deployment managers and engineers to track releases and view component metadata, integrating Jenkins, GitLab, Jira, and AWS (S3, Cognito, STS, QuickSight, DynamoDB) with real-time WebSocket updates and custom D3 visualisations.
- Rebuilt two React applications for Solutions Architects, translating Figma designs into production-ready interfaces with interactive D3 visualisations and a custom JSON editor built with CodeMirror.
- Championed the adoption of and/or migration to TypeScript and Redux Toolkit across our applications.
- Worked across several core applications, addressing bugs, developing features, reviewing code, and producing documentation.
- Drove frontend decision-making by authoring business cases and impact assessments for initiatives including an Axios-to-native-Fetch migration and adoption of updated security standards, bridging engineering rationale and business outcomes.
- Mentored new hires through onboarding, project delegation, and development queries - setting SMART goals, writing formal progress reports, and co-developing training pathways.

Data Analyst

Contentsquare | 2018 - 2019

- Analysed large volumes of web analytics data using SQL, Python, and proprietary software to surface actionable UX insights for enterprise clients, with a focus on understanding user behaviour to improve conversion and retention.
- Identified and independently resolved a manual data extraction bottleneck by automating the population of client-facing Google Sheets and Data Studio dashboards in Python on a cron schedule, ensuring deliverables were ready before the working day began.
- Led workshops for cross-functional stakeholders including software developers, product managers, and UI specialists, translating complex data findings into concrete recommendations.

PhD Researcher

University College London (BBSRC London Interdisciplinary Doctoral Programme) | 2014 - 2018

Supervised by Professor Mark Miodownik (UCL Faculty of Engineering) and Dr Brian Stramer (Randall Centre for Cell & Molecular Biophysics, KCL).

- Developed novel computational techniques in MATLAB and Python to analyse immune cell migration, resulting in a first-author publication in Nature Cell Biology (2019).
- Contributed research ideas, methods, and tools to further work published in Current Biology, Nature Communications, and the Journal of Cell Science.
- Collaborated closely with engineers, geneticists, and cell biologists as a core part of my interdisciplinary research approach.
- Invited to present at the 8th Abercrombie Meeting, a Royal Microscopical Society event held every five years to showcase major advances in cell biology.

EDUCATION

University College London 2014 - 2018 BBSRC London Interdisciplinary Doctoral Programme: Interdisciplinary research alongside engineers, cell biologists, and material scientists. Core thesis published in Nature Cell Biology.

University of Kent 2013 - 2014 MSc Physics (Research): Astrobiology research project published in the Proceedings of the Royal Society.

University of Kent 2010 - 2013 BSc Hons Biomedical Sciences, 2:1: Modules included Immunology, Genetics, and Statistics.

TECHNICAL SKILLS

Languages & Frameworks

JavaScript / TypeScript, React, Node.js, HTML, CSS, Next.js, Python, Go, Ruby, Rails, Java, SQL, NoSQL, MATLAB, R

Tools & Platforms

Git (GitLab, GitHub), Jenkins, Jira / Confluence, Figma, Jest, AWS

General Skills

Project management, strategic thinking, technical communication, data analysis, data visualisation, research

ADDITIONAL TRAINING

Flatiron School 2019 Full Stack Software Engineer: HTML, CSS, JavaScript, and Ruby, culminating in a capstone project built in React with a Ruby on Rails backend.

INTERESTS

Interested in science, technology, and entrepreneurship, with a focus on the intersection of research and innovation. Previously co-hosted and edited the *Radical Science Podcast*, interviewing innovators across open science, startup founders, and metascience researchers. Built a newsletter reaching ~2,500 subscribers and managed a community of ~800 scientists, technologists, and entrepreneurs. Organised events on topics ranging from biotech to quantum computing, bringing together startups, academics, and industry leaders. For several years I was an Innovation Jury Member for the SXSW Innovation Awards. Outside of that, I enjoy being with my two young children, reading, running trails, swimming, and strength training. I've recently picked up Jiu Jitsu and gravel cycling, and am looking to get back into ice hockey after a few years away from the rink.

PUBLICATIONS

AL Law et al. Nance-Horan Syndrome-like 1 protein negatively regulates Scar/WAVE-Arp2/3 activity and inhibits lamellipodia stability and cell migration. Nature Communications, 2021.

L. Yolland. A Destruction/Contraction Gradient Coordinates a Persistent and Polarized Global Actin Flow to Control Cell Directionality. Doctoral thesis (Ph.D), UCL, 2019.

L Yolland et al. Persistent and polarized global actin flow is essential for directionality during cell migration. Nature Cell Biology, 2019.

S Brayford et al. Heterotypic contact inhibition of locomotion can drive cell sorting between epithelial and mesenchymal cell populations. Journal of Cell Science, 2019.

BJ Sanchez-Sanchez et al. A moving source of matrix components is essential for de novo basement membrane formation. Current Biology, 2017.